Teacher's Signature:____________
Date: ________________________

United International **University**

Name: ____ ID: ______________

Section: ______ Batch: _____________ Date: ________________________

Experiment No. 04

Name of the Experiment: Determination of the spring constant and effective mass of a given spiral spring.

Theory:

If a spring is clamped vertically at the end P, and loaded with a mass m_0 at the other end A, then the period of vibration of the spring along a vertical line is given by,

$$T = 2\pi\sqrt{\frac{m' + m_0}{k}} = 2\pi\sqrt{\frac{M}{k}} \quad \ldots\ldots\ldots (1)$$

Where, m' is a constant called the effective mass of the spring and k, the spring constant i.e. the ratio between the added force and the corresponding stretch of the spring.

The contribution of the mass of the spring to the effective mass of the vibrating system can be shown as follows: Consider the kinetic energy of a spring and its load undergoing simple harmonic motion. At the instant under consideration, let the load m_0 be moving with velocity v_0 as shown in the figure.

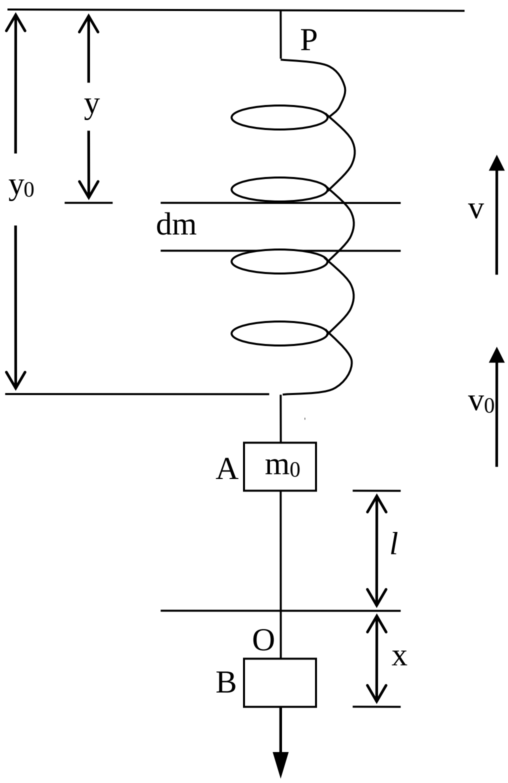

Fig. 01

At the same instant, an element dm of the mass m of the spring will also be moving up but with a velocity v which is smaller than v_0. It is evident that the ratio between v and v_0 is just the ratio between y and y_0. Hence, $\frac{v}{y} = \frac{v_0}{y_0}$ i.e. $v = \frac{v_0}{y_0} y$ .The kinetic energy of the spring alone will be $\int \frac{v^2}{2} dm$. But dm maybe written as $\frac{m}{y_0} dy$, where, m is the mass of the spring.

Thus the integral equals to $\frac{1}{2}\left(\frac{m}{3}\right)v_0^2$. The total kinetic energy of the system will then be $\frac{1}{2}\left(m_0 + \frac{m}{3}\right)v_0^2$ and the total mass of the system is therefore, $M = \left(m_0 + \frac{m}{3}\right)$

Hence, effective mass, $m' = \frac{m}{3}$. The applied force $m_0 g$ is proportional to the extension l within the elastic limit.

$$\text{Therefore, } mg = kl \text{ or, } l = \frac{g}{k} m$$

$$k = \frac{g}{l/m} = \frac{g}{\text{slope of l vs m graph}} =$$

Apparatus:

- A spiral spring
- Convenient masses with hanging arrangement
- Clamp or a hook attached to a rigid framework of heavy metal rods
- Weighing balance
- Stopwatch and scale

Experimental Data:

(A) Initial length of the Spring, L_0 =

(B) Table for determining extensions and time periods:

No. of Obs.	Added Loads, m_0 (gm.)	Length of the Spring, L (cm)	Extension, $l = L_0 - L$ (cm.)	No. of vibrations	Total time (sec.)	Period, T (sec.)	T^2
1	50			20			
2	100			20			
3	150			20			
4	200			20			
5	250			20			

Calculation:

(A) Effective mass, $m^{/}$ (from graph) =

(B) Spring Constant $k = \frac{g}{\text{slope of l vs m graph}} =$

(C) Difference = [(Experimental Result ~ Theoretical Result)/Theoretical Result] x100%
= = =

(D) Accuracy= 100% - % Difference =

Results:

(A) Value of the Effective Mass, m' =

(B) Value of the Spring Constant, k =

Discussions:

Q: What do you understand by the term *Spring Constant*?

Q: What is the *Effective Mass* of a spring?

Q: Infer the extension of your spring when a load of 300 gm is used from the load ~ extension curve.

Q: Does the time period of oscillation depends on the displacement from the equilibrium position? Explain

Q: What happens to the time period if you keep increasing the load?

Q: What type of motion does the spring-block system have? Write the differential equation for such motion. Write the solution of the differential equation.

Q: Suppose two spring with the same spring constant *k*. Draw the diagram, when these two springs are in (i)series and (ii) parallel combination.

Show that the equivalent spring constant

(i) $k_{series} = \frac{k}{2}$

(ii) $k_{parallel} = 2k$